CUSTOMER I/O LIST - MV EXAMPLE

Prepared for the automation contractor. Fictional customer, vessel and data (Bosun sample project).

This list covers the machinery spaces, the boiler plant, power management and the ballast water treatment system.

Pages 2, 3 and 5 are exported from the customer's engineering database. Page 4 is a scanned copy of a hand-corrected page.

Signal types: AI analog input, AO analog output, DI digital input, DO digital output. Alarm limits are in engineering units.

Delays are given in seconds. Blank cells mean not applicable.

Revision B, issued for review. Please return comments within two weeks.

I/O LIST - MACHINERY SPACE

Exported table

Tag	Description	Type	Unit	Range low	Range high	HH	H	L	LL	Delay s
TANK-TK25-LT02	Slop tank 25 level 2	AI	%	0	100	80	88	25	5	0
TANK-TK25-LT03	Slop tank 25 level 3	AI	%	0	100	88	85	15	11	15
TANK-TK25-TT01	Slop tank 25 temperature 1	AI	°C	0	100		80			30
TANK-TK25-TT02	Slop tank 25 temperature 2	AI	°C	0	100		85			5
TANK-TK25-PT1	Slop tank 25 vapour pressure	AI	mbar	-50	150		120	-10		10
TANK-TK26-LT01	Fresh water tank 26 level 1	AI	%	0	100		75	25	5	30
TANK-TK26-LT02	Fresh water tank 26 level 2	AI	%	0	100		85	15	11	15
TANK-TK26-LT03	Fresh water tank 26 level 3	AI	%	0	100		75	20	5	2
TANK-TK26-LT04	Fresh water tank 26 level 4	AI	%	0	100		80	15	5	0
TANK-TK26-TT01	Fresh water tank 26 temperature	AI	°C	0	60		48			0
TANK-OWS2-AT01	Oil content monitor 1 (bank 2)	AI	ppm	0	30		24			3
TANK-OWS2-AT02	Oil content monitor 2 (bank 2)	AI	ppm	0	30		24			10
TANK-OWS2-LT01	Bilge holding tank level 1 (bank 2)	AI	%	0	100	96	75	20		5
TANK-OWS2-LT02	Bilge holding tank level 2 (bank 2)	AI	%	0	100	88	75	15		2
TANK-OWS2-FT01	Separator discharge flow (bank 2)	AI	m3/h	0	5		4			10
TANK-OWS2-XA01	Separator alarm 1 (bank 2)	DI								
TANK-OWS2-XA02	Separator alarm 2 (bank 2)	DI								
ME-HPU2-PT01	Hydraulic power unit pressure 1 (bank 2)	AI	bar	0	400		340	60	32	5
ME-HPU2-PT02	Hydraulic power unit pressure 2 (bank 2)	AI	bar	0	400		320	60	44	10
ME-HPU2-PT03	Hydraulic power unit pressure 3 (bank 2)	AI	bar	0	400		320	100	44	0
ME-HPU2-TT01	Hydraulic oil temperature 1 (bank 2)	AI	°C	0	100	92	85			10
ME-HPU2-TT02	Hydraulic oil temperature 2 (bank 2)	AI	°C	0	100	92	85			15
ME-HPU2-LT01	Hydraulic oil tank level (bank 2)	AI	%	0	100			25	11	5
ME-HPU2-XS01	Hydraulic pump 1 running (bank 2)	DI								
ME-HPU2-XS02	Hydraulic pump 2 running (bank 2)	DI								
ME-HPU2-XS03	Hydraulic pump 3 running (bank 2)	DI								
ME-HPU2-XS04	Hydraulic pump 4 running (bank 2)	DI								
ME-EXH-TT03	Cylinder pressure indicator 1 (bank 2)	AI	bar	0	200		150			15

I/O LIST - MACHINERY SPACE (CONT.)

Exported table

Tag	Description	Type	Unit	Range low	Range high	HH	H	L	LL	Delay s
ME-CPD2-PT02	Cylinder pressure indicator 2 (bank 2)	AI	kgf/cm2	0	200		160			10
ME-CPD2-PT03	Cylinder pressure indicator 3 (bank 2)	AI	bar	0	200		150			10
ME-CPD2-PT04	Cylinder pressure indicator 4 (bank 2)	AI	bar	0	200		160			15
ME-CPD2-TT01	Cylinder liner temperature 1 (bank 2)	AI	°C			264	225			15
ME-CPD2-TT02	Cylinder liner temperature 2 (bank 2)	AI	°C	0	300	276	225			10
ME-CPD2-TT03	Cylinder liner temperature 3 (bank 2)	AI	°C	0	300	264	240			5
ME-CPD2-TT04	Cylinder liner temperature 4 (bank 2)	AI	°C	0	300	276	255			0
ME-CPD2-VT01	Cylinder block vibration 1 (bank 2)	AI	mm/s	0	50	44	42.5			5
ME-CPD2-VT02	Cylinder block vibration 2 (bank 2)	AI	mm/s	0	50	48	40			15
AE-SCV2-TT01	Scavenge air temperature 1 (bank 2)	AI	°C	0	150	144	128			10
AE-SCV2-TT02	Scavenge air temperature 2 (bank 2)	AI	°C	0	150	132	112.5			10
AE-SCV2-TT03	Scavenge air temperature 3 (bank 2)	AI	°C	0	150	132	127.5			3
AE-SCV2-PT01	Scavenge air pressure 1 (bank 2)	AI	bar	0	4		3.4	0.8		5
AE-SCV2-PT02	Scavenge air pressure 2 (bank 2)	AI	bar	0	4		3.2	0.8		5
AE-SCV2-PT03	Scavenge air pressure 3 (bank 2)	AI	bar	0	4		3.2	0.6		15
AE-SCV2-ST01	Turbocharger speed 1 (bank 2)	AI	rpm	0	40000	36800	30000			0
AE-SCV2-ST02	Turbocharger speed 2 (bank 2)	AI	rpm	0	40000	38400	32000			15
BLR-SPH2-TT01	Steam superheater outlet temperature 1 (bank 2)	AI	°C	0	450	396	382			30
BLR-SPH2-TT02	Steam superheater outlet temperature 2 (bank 2)	AI	°C	0	450	432	337.5			0
BLR-SPH2-TT03	Steam superheater outlet temperature 3 (bank 2)	AI	°C	0	450	414	337.5			10
BLR-SPH2-PT01	Steam header pressure 1 (bank 2)	AI	bar	0	25	24	18.8	3.8		3
BLR-SPH2-PT02	Steam header pressure 2 (bank 2)	AI	bar	0	25	22	18.8	6.2		3
BLR-SPH2-PT03	Steam header pressure 3 (bank 2)	AI	bar	0	25	23	18.8	3.8		0
BLR-SPH2-FT01	Steam flow 1 (bank 2)	AI	t/h	0	40.5		32.4			5
BLR-SPH2-FT02	Steam flow 2 (bank 2)	AI	t/h	0	40.5		30.4			30
PMS-UPS2-VT01	UPS battery voltage 1 (bank 2)	AI	V	0	300		255	45	24	15
PMS-UPS2-VT02	UPS battery voltage 2 (bank 2)	AI	V	0	300		225	75	33	5
PMS-UPS2-VT03	UPS battery voltage 3 (bank 2)	AI	V	0	300		240	75	33	10

I/O LIST - BOILER PLANT

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Tag	Description	Type	Unit	Range low	Range high	HH	H	L	LL	Delay s
PMS-UPS2-II01	UPS output current 1 (bank 2)	AI	A	200	0		150			30
PMS-UPS2-II02	UPS output current 2 (bank 2)	AI	A	0	200		160			10
PMS-UPS2-II03	UPS output current 3 (bank 2)	AI	A	0	200		170			15
PMS-UPS2-TT01	UPS cabinet temperature 1 (bank 2)	AI	°C	0	80	96	60			5
PMS-UPS2-TT02	UPS cabinet temperature 2 (bank 2)	AI	°C	0	80	73.6	64			0
PMS-UPS2-XA01	UPS alarm 1 (bank 2)	DI								
PMS-UPS2-XA02	UPS alarm 2 (bank 2)	DI								
PMS-UPS2-XA03	UPS alarm 3 (bank 2)	DI								
BWTS-CHM2-LT01	Neutralising chemical tank level 1 (bank 2)	AI	%	0	100			25	11	10
BWTS-CHM2-LT02	Neutralising chemical tank level 2 (bank 2)	AI	%	0	100			20	5	5
BWTS-CHM2-FT01	Chemical dosing flow 1 (bank 2)	AI	L/h	0	250		200		27.5	30
BWTS-CHM2-FT02	Chemical dosing flow 2 (bank 2)	AI	L/h	0	250		212		12	0
BWTS-CHM2-PT01	Chemical dosing pressure 1 (bank 2)	AI	bar	0	10		8			2
BWTS-CHM2-PT02	Chemical dosing pressure 2 (bank 2)	AI	bar	0	10		8.5			2
BWTS-CHM2-YS01	Dosing pump 1 start (bank 2)	DO								
BWTS-CHM2-YS02	Dosing pump 2 start (bank 2)	DO								
ME-SCR3-TT01	SCR catalyst bed 1 temperature (bank 3)	AI	°C	0	500	480	375			0
ME-SCR3-TT02	SCR catalyst bed 2 temperature (bank 3)	AI	°C	0	500	440	375			3
ME-SCR3-TT03	SCR catalyst bed 3 temperature (bank 3)	AI	°C	0	500	440	375			5
ME-SCR3-TT04	SCR catalyst bed 4 temperature (bank 3)	AI	°C	0	500	460	400			30

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I/O LIST - POWER MANAGEMENT

Exported table

Tag	Description	Type	Unit	Range low	Range high	HH	H	L	LL	Delay s
ME-SCR3-PT01	SCR catalyst differential pressure 1 (bank 3)	AI	mbar	0	300		255			2
ME-SCR3-PT02	SCR catalyst differential pressure 2 (bank 3)	AI	mbar	0	300		225			3
ME-SCR3-AT01	NOx analyser reading 1 (bank 3)	AI	ppm	0	2000	1760	1700			5
ME-SCR3-AT02	NOx analyser reading 2 (bank 3)	AI	ppm	0	2000	1920	1500			5
ME-SCR3-FT01	Urea dosing flow (bank 3)	AI	L/h	0	120			30	6	2
ME-SCR3-XS01	Urea pump 1 running (bank 3)	DI								
ME-SCR3-XS02	Urea pump 2 running (bank 3)	DI								
ME-SCR3-XS03	Urea pump 3 running (bank 3)	DI								
ME-EGR3-TT01	EGR cooler outlet temperature 1 (bank 3)	AI	°C	0	400	384	320			5
ME-EGR3-TT02	EGR cooler outlet temperature 2 (bank 3)	AI	°C	0	400	352	300			5
ME-EGR3-TT03	EGR cooler outlet temperature 3 (bank 3)	AI	°C	0	400	384	320			5
ME-EGR3-PT01	EGR scrubber pressure 1 (bank 3)	AI	bar	0	5		4	0.75		5
ME-EGR3-PT02	EGR scrubber pressure 2 (bank 3)	AI	bar	0	5		4	0.75		10
ME-EGR3-ST01	EGR blower speed (bank 3)	AI	rpm	0	12000	10560	9600			3